

Flujo máximo y corte mínimo

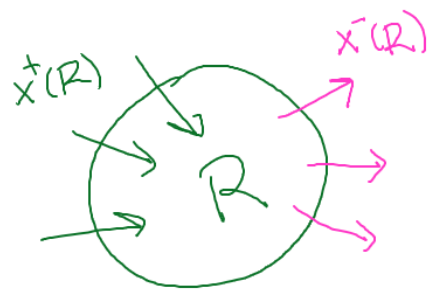
Flujo

• Digrafo D

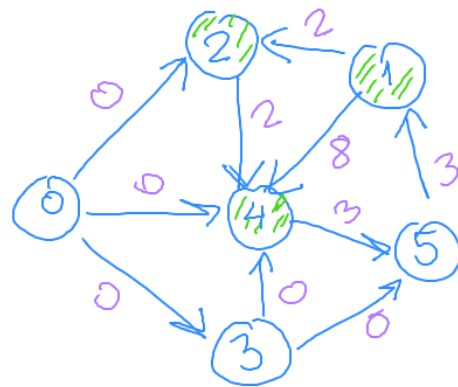
• flujo: función de $E(D)$ hacia $\mathbb{R}^+$ // $x \in E(D) \rightarrow \mathbb{R}^+$

• flujo x , $R \subseteq V(D)$, denotamos $x^+(R), x^-(R) \rightarrow \sum_{\partial^+(R)} x_{uv}$

$$\begin{cases} \partial^+(R) = \{uv \mid u \in R, v \notin R\} \\ \partial^+(V) = \emptyset \end{cases}$$



E_m



$$\begin{aligned} x^+(R) &= 3 + 0 + 0 \\ x^-(R) &= 3 \end{aligned} \quad \text{) } ex_m = 0$$

$$x_{-}^+(R) = x^+(R) - x^-(R)$$

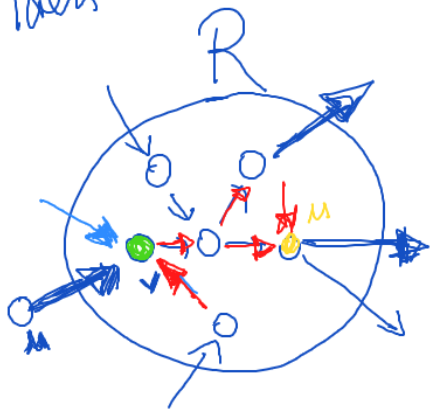
exceso

Obs $x^-(R) = x^+(\bar{R})$
 $\uparrow$
 $V \setminus R$

Lemma 1

$$x^{\pm}(R) = \sum_{u \in R} x^{\pm}(u)$$

P/ Idea

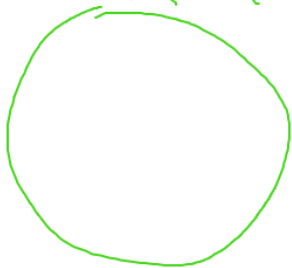


$$x^+(R) - x^-(R) = \sum_{uv \in \partial^+(R)} x_{uv} - \sum_{uv \in \partial^-(R)} x_{uv}$$

$$= \sum_{\substack{uv: u \in \bar{R}, v \in R}} x_{uv} - \sum_{u \in R, v \in \bar{R}} x_{uv} + \sum_{uv: u \in R, v \in \bar{R}} x_{uv} - \sum_{uv: u \in \bar{R}, v \in R} x_{uv}$$

$$= \sum_{v \in R} x^+(v) - \sum_{u \in R} x^-(u) //$$

$$R = V(D)$$



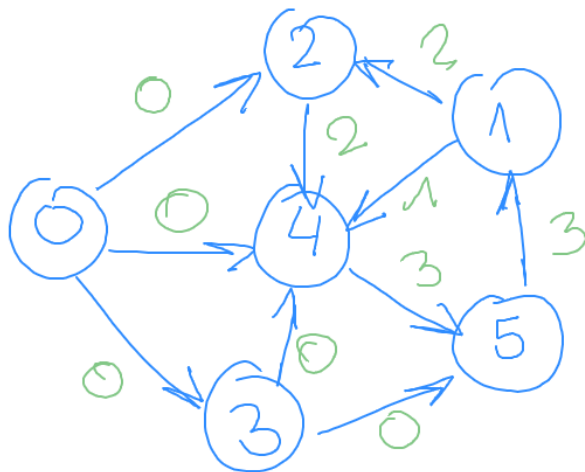
$$x^+(R) = x^-(R) = 0$$

Circulación

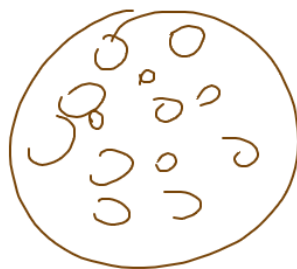
$$x_{-}^{+}(u) = 0$$

para todo $u \in V(D)$.

Ejm



Obs $x^{+}(R) = x^{-}(R) \quad \forall R \subseteq V(D)$



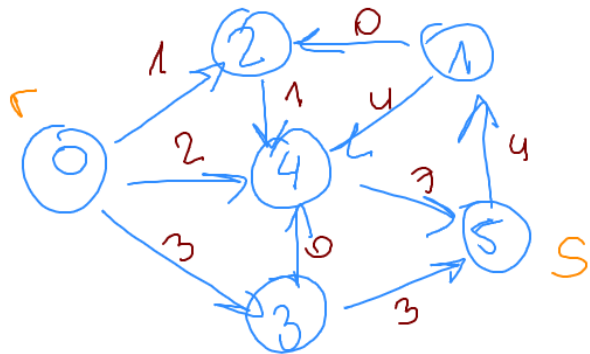
Flujo entre 2 nodos

- $r, s \in V(D)$

- flujo de r a s (rs-flujo) cumple que

$$x_{-}^{+}(u) = \begin{cases} 0 & ; u \notin \{r, s\} \\ \geq 0 & ; u \in S \end{cases}$$

- Valor de x , es $f(x) = x_{-}^{+}(s)$.



$$\underline{f(x) = 6}$$

Obs $f(x) = x_{-}^{+}(r)$

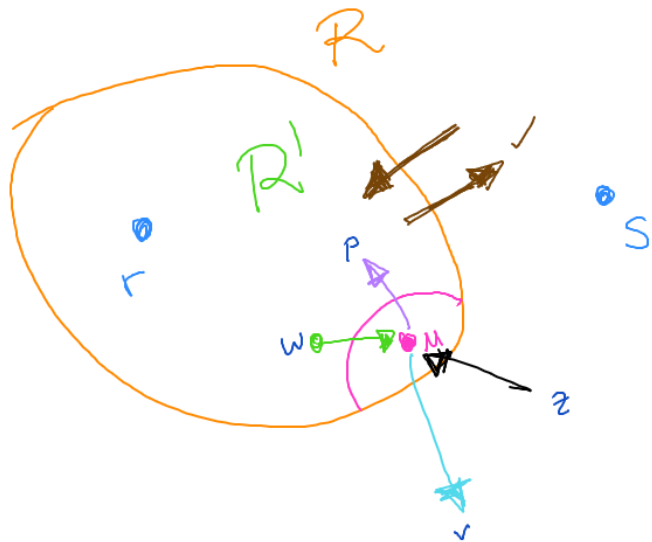
Lemma 2

x is flow of r to s

$R \subseteq V(D)$ tq $r \in R, s \notin R. \Rightarrow f(x) = x_+^-(R)$

$$\Leftrightarrow \begin{cases} x_+^-(R) = -x_+^+(R) \end{cases}$$

Prueba (Idea: induccion) $\rightarrow$ en R



• $|R| = 1, R = \{r\}$

$$f(x) = x_+^-(r) = x_+^-(R) \checkmark$$

• $|R| > 1, R' = R - u$

Por h.o.p., $f(x) = x_+^-(R')$

$$x_+^-(R) = x_+^-(R') + \sum_{uv: v \in R} x_{uv} - \sum_{wu: w \in R} x_{wu}$$

$$x_+^+(R) = x_+^+(R') + \sum_{zu: z \in R} x_{zu} - \sum_{up: p \in R} x_{up}$$

$$x_+^-(R) = x_+^-(R) - x_+^+(R) = x_+^-(R') + x_+^+(u) - x_+^-(u) = x_+^-(R') = f(x)$$

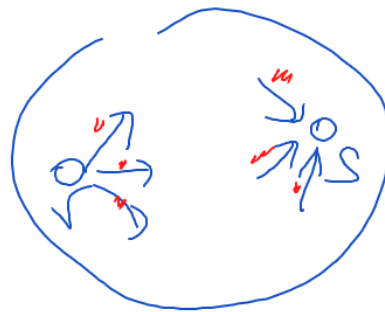
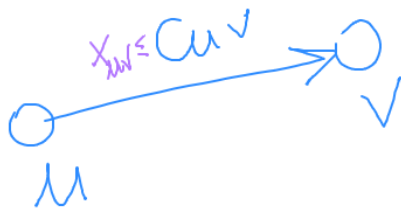
Flujo máximo

Sea $c: E(D) \rightarrow \mathbb{R}^+$ // Capacidad

x respeta c si $x_{uv} \leq c_{uv} \quad \forall uv \in E(D)$.

↑ (flujo viable)

Problema de flujo máximo: encontrar flujo viable de r a s de valor máximo



Cortes

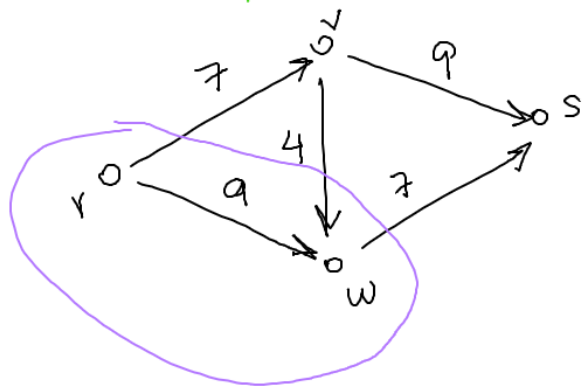
- $R \subseteq V(D)$

- $c(R) = c(\partial^-(R)) = \sum_{uv \in \partial^-(R)} \text{Curv}$

↑ capacidad del corte

- $\partial^-(R)$ separa r de s si $r \in R, s \notin R$ o También rs -corte

Ejm



$$c(\{r\}) = 7 + 9 = 16$$

$$c(r, v) = 22$$

$$c(r, w) = 7 + 7 = 14$$

$$c(r, v, w) = 16.$$



Lema 3

x : rs-flyoviable

$\partial^-(R)$: rs-corte

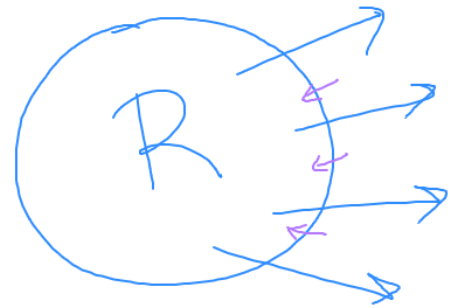


$$f(x) \leq c(R).$$

P/ Por Lema 2, $f(x) = x^+(R)$. Además, como x es viable
 $x^-(R) \leq c(R)$

Luego,

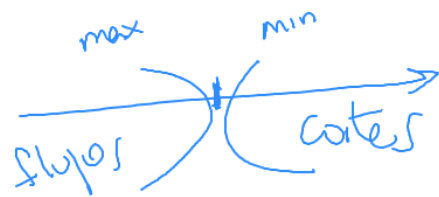
$$\begin{aligned} f(x) &= \underline{x^-(R)} - x^+(R) \\ &\leq \underline{c(R)} - x^+(R) \\ &\leq c(R) \end{aligned}$$



Teorema de flujo máximo y corte mínimo

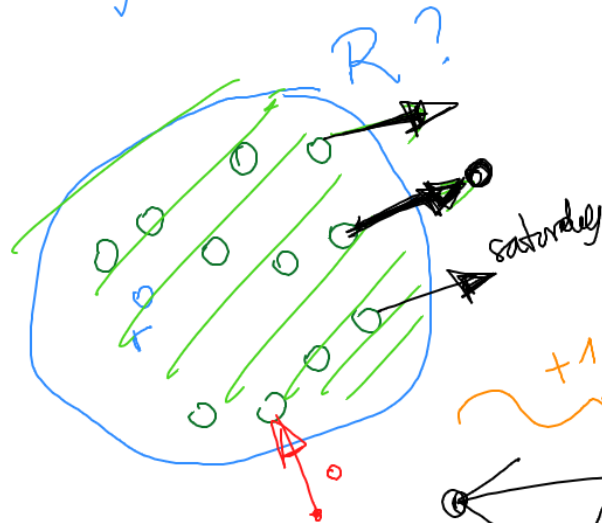
Existe un flujo viable x de r a s
 y un r - s corte $\partial(P)$ tq $f(x) = c(P)$

...

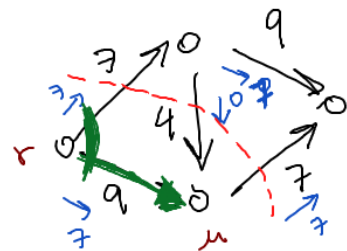


idem

¿hay flujo máximo x (de r a s)



NO HAY CAM
aument de r a s



$$R = \{v \in V \mid \exists \text{ cam } r \rightarrow v \text{ en } G\}$$

Cam aumentador

